I Tangents and Normals

Investigation

The curve $y = x^3 - 4x$ is shown alongside.

Things to think about:

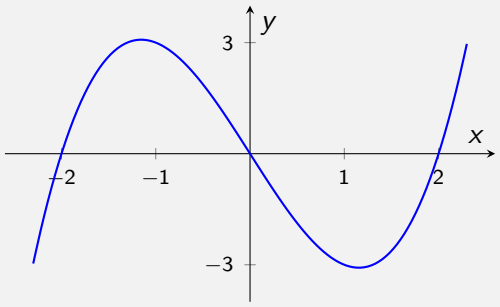
a) At which points is the tangent to the curve horizontal?

b) On which intervals is the curve:

i) increasing

ii) decreasing

c) At which point does the curve change its *shape*? How can we use calculus to find this point?



Property

Consider a curve $y = f(x)$. If A is the point with x-coordinate a, then the gradient of the tangent to the curve at this point is

$$m_T = f'(a).$$

The equation of the tangent is

$$\frac{y - f(a)}{x - a} = f'(a) \quad \text{or} \quad y = f'(a)(x - a) + f(a).$$

Definition

A **normal** to a curve is a line which is perpendicular to the tangent at the point of contact.

Property

If $f'(a) \neq 0$, the gradient of the **normal** to the curve at $x = a$ is

$$m_N = -\frac{1}{f'(a)}.$$

(If $f'(a) = 0$ the tangent is horizontal, so the normal is the vertical line $x = a$.)

Example

If $f(x) = x^2$ then $f'(x) = 2x$. At $x = 2$,

$$m_T = f'(2) = 4 \quad \text{and} \quad m_N = -\frac{1}{f'(2)} = -\frac{1}{4}.$$

Since $f(2) = 4$, the tangent has equation

$$y - 4 = 4(x - 2) \quad \text{or} \quad y = 4x - 4,$$

and the normal has equation

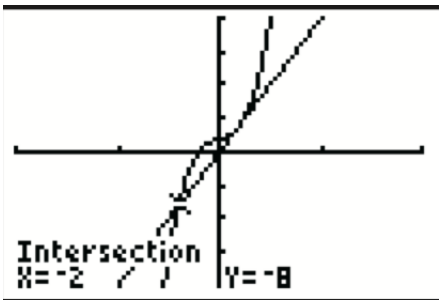
$$y - 4 = -\frac{1}{4}(x - 2) \quad \text{or} \quad y = -\frac{1}{4}x + \frac{9}{2}.$$

Example

Find the equation of the tangent to $f(x) = x^2 + 1$ at the point where $x = 1$.

Example

Find the coordinates of the point where the tangent to $y = x^3 + x + 2$ at $(1, 4)$ meets the curve again.



Example

Find the equations of the tangents to $y = x^2$ from the external point $(2, 3)$.

EXERCISE 1

Find the equations of the tangent and of the normal to $f(x) = x^3 - 3x + 1$ at the point where $x = 2$.

EXERCISE 2

Find the equation of the tangent to $y = x^2 - 4x + 7$ which is parallel to the line $y = 2x + 1$.

EXERCISE 3

The tangent to $y = x^3 - 2x$ at $(-1, 1)$ meets the curve again at a point P. Find the coordinates of P.

EXERCISE 4

Find the equations of the tangents to $y = x^2$ from the external point $(1, -3)$.

EXERCISE 5

Find the equation of the normal to $y = e^{2x}$ at the point where $x = 0$.

II Increasing and Decreasing Functions

Property

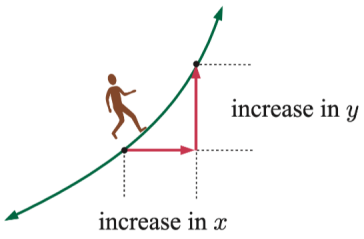
Suppose S is an interval in the domain of $f(x)$, so $f(x)$ is defined for all x in S.

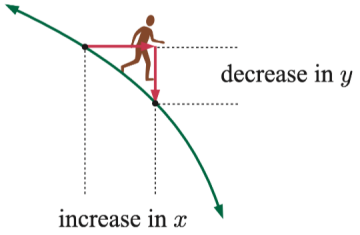
- $f(x)$ is **increasing** on $S \Leftrightarrow$ $f(a) \leq f(b)$ for all $a, b \in S$ such that $a < b$.
- $f(x)$ is **decreasing** on $S \Leftrightarrow$ $f(a) \geq f(b)$ for all $a, b \in S$ such that $a < b$.

Property

Suppose f is differentiable on an interval S.

- $f(x)$ is **increasing** on $S \Leftrightarrow$ $f'(x) \geq 0$ for all x in S.
- $f(x)$ is **decreasing** on $S \Leftrightarrow$ $f'(x) \leq 0$ for all x in S.





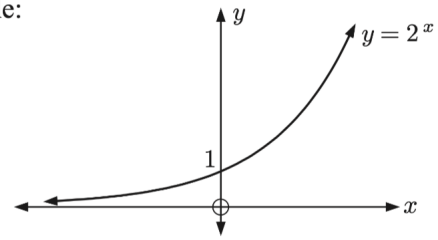
Remark.

- $f(x)$ is **strictly increasing** on S if $f(a) < f(b)$ whenever $a < b$ in S. This is guaranteed if $f'(x) > 0$ for all x in S.
- $f(x)$ is **strictly decreasing** on S if $f(a) > f(b)$ whenever $a < b$ in S. This is guaranteed if $f'(x) < 0$ for all x in S.
- The converse is false: $f(x) = x^3$ is strictly increasing, yet $f'(0) = 0$.

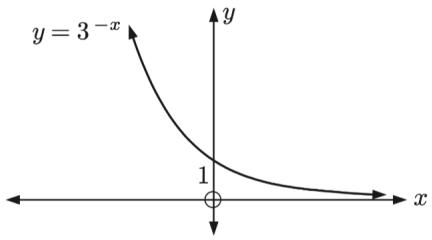
Definition

When a function is either increasing or decreasing for all $x \in \mathbb{R}$, we say it is **monotone increasing** or **monotone decreasing**.

For example:



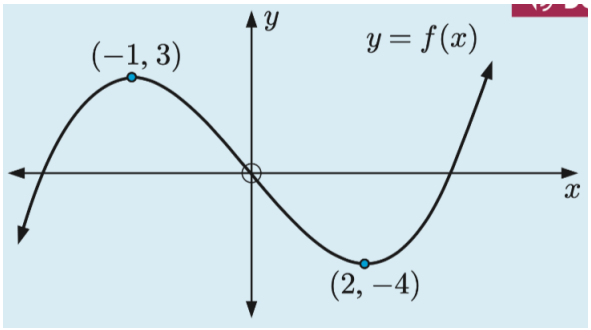
$y = 2^x$ is strictly increasing for all x.



$y = 3^{-x}$ is strictly decreasing for all x.

Example

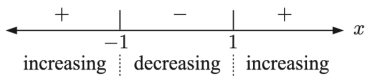
Find intervals where $f(x)$ is: a) increasing b) decreasing.



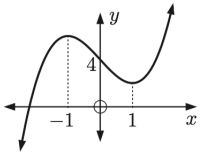
Sign diagrams for the derivative are extremely useful for determining intervals where a function is increasing or decreasing. Consider the following example:

$$\begin{aligned} f'(x) &= 3x^2 - 3 \\ &= 3(x^2 - 1) \\ &= 3(x+1)(x-1) \end{aligned}$$

which has sign diagram



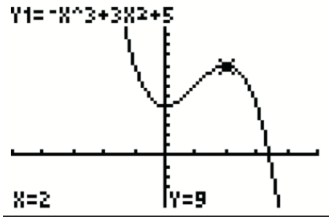
$f(x) = x^3 - 3x + 4$



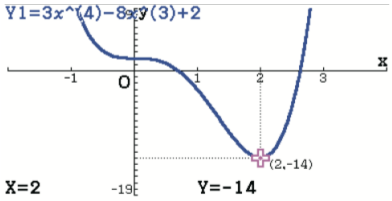
Example

Find the intervals where the following functions are increasing or decreasing:

a) $f(x) = -x^3 + 3x^2 + 5$



b) $f(x) = 3x^4 - 8x^3 + 2$



EXERCISE 6

Find the intervals on which each function is increasing or decreasing.

a) $f(x) = x^3 - 6x^2 + 9x$

b) $f(x) = x^4 - 2x^2$

c) $f(x) = x\,e^{-x}$

EXERCISE 7

Show that $f(x) = x^3 + 3x + 1$ is monotone increasing.

EXERCISE 8

Find all values of k for which $f(x) = x^3 + kx^2 + 3x$ is increasing for all $x \in \mathbb{R}$.

III Stationary Points

a Extrema

Definition

A **stationary point** of a function is a point where $f'(x) = 0$. It could be a **local maximum**, a **local minimum**, or a **stationary inflection**.

Remark.
- A **global minimum** is the minimum value of y on the entire domain.
- A **global maximum** is the maximum value of y on the entire domain.

Example

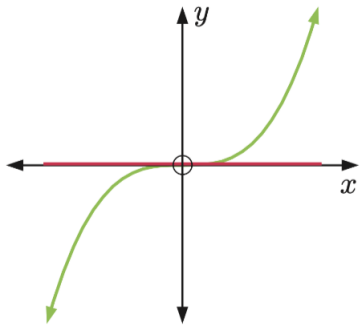
For $y = x^2$, the point $(0, 0)$ is a local minimum and is also the global minimum.

b Stationary points of inflection

It is not always true that whenever we find a value of x where $f'(x) = 0$, we have a local maximum or minimum.

Example

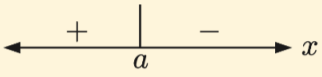
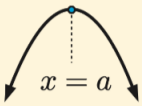
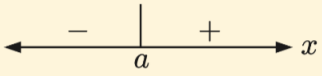
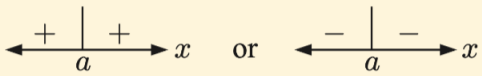
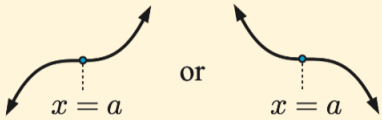
$f(x) = x^3$ has $f'(x) = 3x^2$ and $f'(x) = 0$ when $x = 0$.



Definition

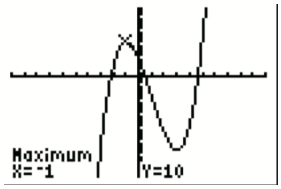
It is called a **stationary inflection** (or **inflexion**) as the curve changes its curvature or shape.

c Sign diagrams

Stationary point where $f'(a) = 0$	*Sign diagram of* $f'(x)$ *near* $x = a$	*Shape of curve near* $x = a$
local maximum		
local minimum		
stationary inflection		

Example

Find and classify all stationary points of $f(x) = x^3 - 3x^2 - 9x + 5$.



Method (Global extrema on a closed interval $a \leq x \leq b$). **Step 1:** Find the stationary points in $[a, b]$.

Step 2: Evaluate f at each stationary point **and** at the endpoints $x = a$ and $x = b$.

Step 3: The largest value is the greatest value, the smallest is the least value.

Example

Find the greatest and least value of $y = x^3 - 6x^2 + 5$ on the interval $-2 \leq x \leq 5$.

EXERCISE 9

Find and classify all stationary points of:

a) $f(x) = 2x^3 - 3x^2 - 12x + 1$

b) $f(x) = x^4 - 4x^3$

c) $f(x) = x + \frac{4}{x}, \quad x \neq 0$

EXERCISE 10

Find the greatest and least value of $f(x) = x^3 - 12x + 5$ on $-3 \leq x \leq 5$.

EXERCISE 11

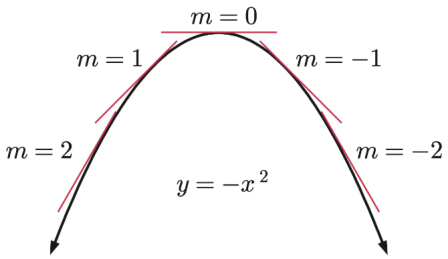
The curve $y = x^3 + ax^2 + bx$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Find a and b.

IV Shape of a Curve

Definition

When a curve, or part of a curve, has shape:

- ∩ we say that the shape is **concave downwards**;
- ∪ we say that the shape is **concave upwards**.



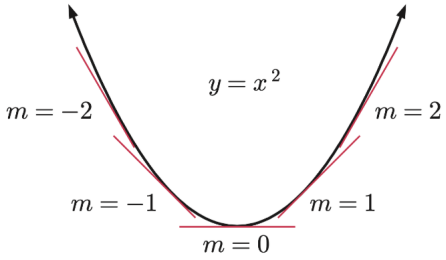
Wherever we are on the curve, as x increases, the gradient of the tangent decreases.

$$\therefore \ f'(x) \text{ is decreasing}$$

$$\therefore \ \text{its derivative is negative,}$$

which means

$$f''(x) < 0.$$



Wherever we are on the curve, as x increases, the gradient of the tangent increases.

$$\therefore \ f'(x) \text{ is increasing}$$

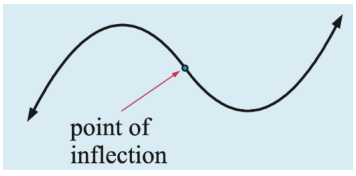
$$\therefore \ \text{its derivative is positive,}$$

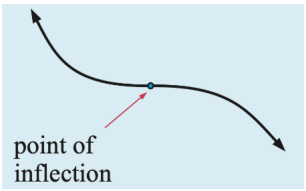
which means

$$f''(x) > 0.$$

Definition

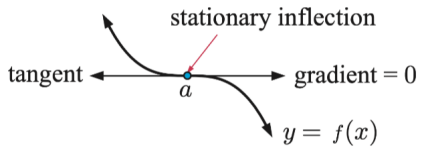
A **point of inflection** is a point on a curve at which there is a change of curvature or shape.





Property

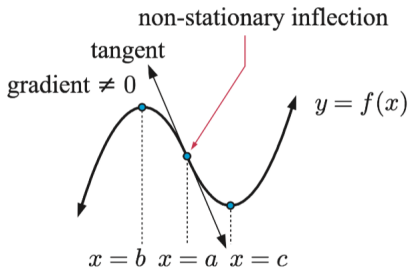
If the tangent at a point of inflection is horizontal, then this point is a **horizontal** or **stationary inflection**.



Sign diagram of $f'(x)$

Sign diagram of $f''(x)$

Remark. If the tangent at a point of inflection is not horizontal, we have a **non-horizontal** or **non-stationary inflection**.



Sign diagram of $f'(x)$

Sign diagram of $f''(x)$

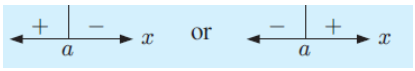
Definition

The tangent at the point of inflection, also called the **inflecting tangent**, crosses the curve at that point.

Property

There is a point of inflection at $x = a$ if $f''(a) = 0$ **and** the sign of $f''(x)$ changes at $x = a$.

The point of inflection corresponds to a change in curvature. In the vicinity of a, $f''(x)$ has sign diagram either



Remark. $f''(a) = 0$ is **not enough**: for $f(x) = x^4$, $f''(0) = 0$ but $f''(x) = 12x^2 \geq 0$ on both sides of 0, so there is no change of concavity and no inflection.

Second derivative test

Suppose $f'(a) = 0$.

- If $f''(a) > 0$, then $(a, f(a))$ is a **local minimum**.
- If $f''(a) < 0$, then $(a, f(a))$ is a **local maximum**.
- If $f''(a) = 0$, the test is **inconclusive**: use a sign diagram of $f'(x)$ (or of $f''(x)$).

Example

Use the second derivative test to classify the stationary points of $f(x) = x^3 - 3x^2 - 9x + 5$.

Example

Find and classify all points of inflection of $f(x) = x^4 - 4x^3 + 5$.

Example

Consider $f(x) = 3x^4 - 16x^3 + 24x^2 - 9$.

a) Find and classify all points where $f'(x) = 0$.

b) Find and classify all points of inflection.

c) Find intervals where the function is increasing or decreasing.

d) Find intervals where the function is concave up or down.

e) Sketch the function showing the features you have found.

Example

Consider the function $y = 2 - e^{-x}$.

a) Find the x-intercept.

b) Find the y-intercept.

c) Show algebraically that the function is increasing for all x.

d) Show algebraically that the function is concave down for all x.

e) Use technology to help graph $y = 2 - e^{-x}$.

f) Explain why $y = 2$ is a horizontal asymptote.

EXERCISE 12

Find the coordinates of the point of inflection of $f(x) = x^3 - 6x^2 + 9x + 1$. State where the curve is concave up and where it is concave down. Is the inflection stationary?

EXERCISE 13

Find all points of inflection of $f(x) = x^4 - 6x^2$ and say whether each is stationary.

EXERCISE 14

Explain why $f(x) = x^4$ has $f''(0) = 0$ but no point of inflection at $x = 0$.

EXERCISE 15

Let $f(x) = x\,e^{-x}$.

a) Show that $f'(x) = (1 - x)e^{-x}$ and $f''(x) = (x - 2)e^{-x}$.

b) Find the stationary point and use the second derivative test to classify it.

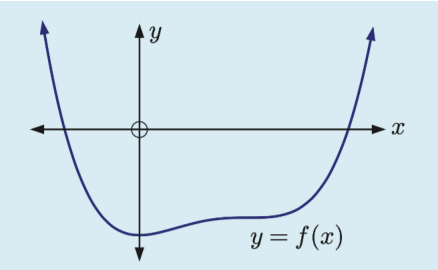
c) Find the point of inflection.

d) Sketch the graph of f, noting that $f(x) \to 0$ as $x \to \infty$.

V Graphs of f, f' and f''

Example

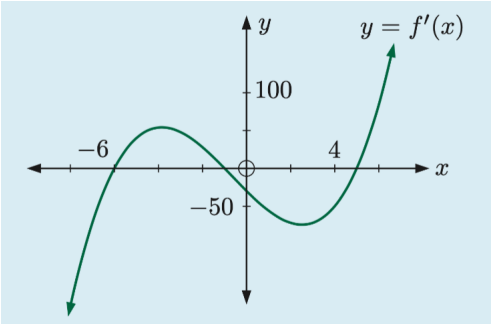
Using the graph of $y = f(x)$ alongside, sketch the graphs of $y = f'(x)$ **and** $y = f''(x)$.



Example

The graph alongside shows a gradient function $y = f'(x)$.

Sketch a graph which could be $y = f(x)$, showing clearly the x-values corresponding to all stationary points and points of inflection.



Summary: reading the graphs

f	f'	f''
increasing	above the x-axis ($f' > 0$)	
decreasing	below the x-axis ($f' < 0$)	
local max / min	crosses the x-axis	$f'' < 0$ / $f'' > 0$
concave up / down	increasing / decreasing	above / below the x-axis
point of inflection	local max / min of f'	crosses the x-axis

EXERCISE 16

The graph of $y = f'(x)$ is a parabola opening upwards which cuts the x-axis at $x = -2$ and $x = 4$.

a) Find the x-values of the stationary points of f and classify them.

b) State the interval(s) on which f is decreasing.

c) Find the x-coordinate of the point of inflection of f and the interval where f is concave up.

EXERCISE 17

A function g satisfies $g'(x) = x^2(x - 3)$. Find the x-values of all stationary points and points of inflection of g, and classify each.